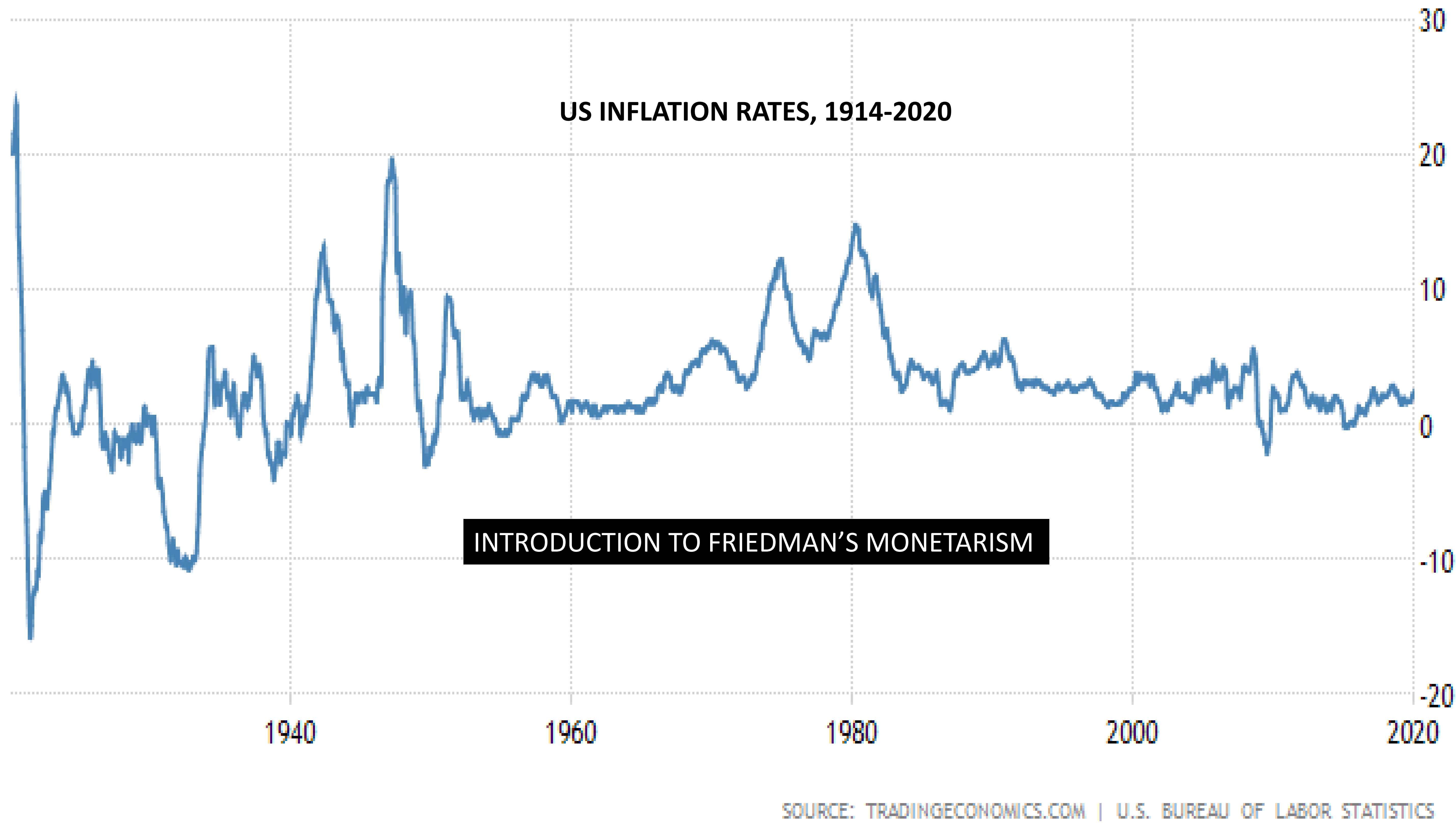
MILTON FRIEDMAN AND MONETARISM

A comprehensive exploration of classical monetary theory and its application to modern central banking policy.



US INFLATION RATES, 1914-2020
INTRODUCTION TO FRIEDMAN'S MONETARISM
30
20
10
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1940
1960
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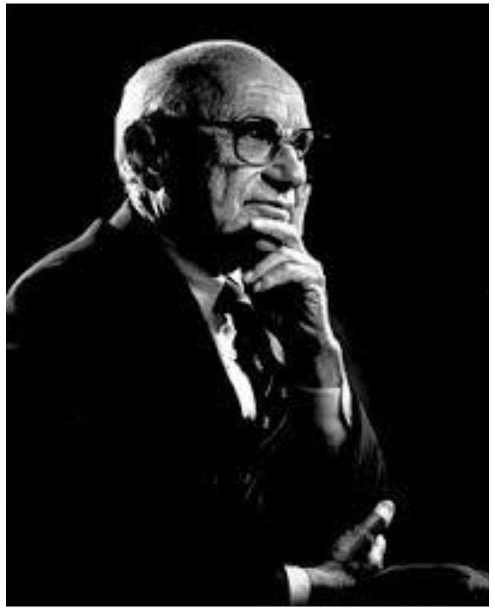
2020
SOURCE: TRADINGECONOMICS.COM | U.S. BUREAU OF LABOR STATISTICS



Milton Friedman, like Keynes (and Walras) was influenced by major macroeconomic events of his time.

As an academic in the 1960s he noticed not only high inflation but also very volatile inflation.

He attributed inflation volatility to erratic monetary policy in a theory that came to be called monetarism

COURSE OVERVIEW

STRUCTURE AND LEARNING RESOURCES

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INTRODUCTION

Context and historical background

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Practical implications for central banks

MILTON FRIEDMAN (1912–2006)

Nobel Prize in Economics, 1976

US economist whose intellectual legacy includes monetarism—a school of thought relating classical monetary theory to modern central bank policy.

Friedman's work fundamentally challenged prevailing Keynesian orthodoxy on monetary policy.

HISTORICAL CONTEXT: THE INFLATION INFLATION PROBLEM

THE CHALLENGE

Constantly fluctuating and often high inflation rates plagued the United States during the 1950s and 1960s.

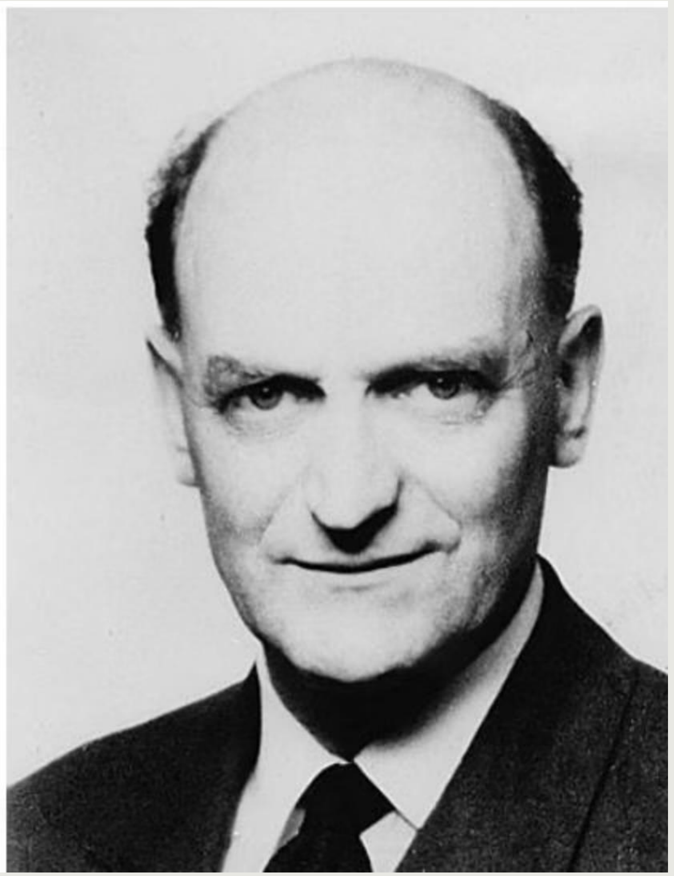
FRIEDMAN'S DIAGNOSIS

Erratic control of the money supply by the Federal Reserve was to blame for unstable inflation.

THE CATALYST

A 1958 article by New Zealand economist A.W. Phillips revealed an empirical phenomenon that would reshape macroeconomic thinking.

A.W. PHILLIPS (1914–1975)



New Zealand economist who discovered an empirical relationship using UK historical data.

Key Finding: A negative relationship existed between the rate of change in wages and the unemployment rate.

THE PHILLIPS CURVE

1

ORIGINAL FINDING

Negative relationship between wage changes and unemployment

2

EXTENDED CONCEPT

Predicted negative relationship between price inflation and unemployment

3

POLICY INTERPRETATION

Promoted as a 'menu of choice' between inflation and unemployment

Keynesian economists, particularly Paul Samuelson, argued that central banks could target specific unemployment rates by engineering corresponding inflation rates through monetary policy.

CRITICAL ANALYSIS

FRIEDMAN'S MONETARIST CRITIQUE

DIRECT CHALLENGE

Friedman's monetarism directly critiqued the Phillips Curve, especially the notion that it represented a policy menu of choice.

POLICY DIAGNOSIS

He blamed reliance on this concept by monetary policymakers for erratic conduct of monetary policy and fluctuating inflation.

LASTING IMPACT

His criticisms are widely credited with the eventual rejection of the Phillips Curve, even by mainstream Keynesian economists.

PART I

LABOUR MARKET UNDERPINNINGS

Classical foundations of macroeconomic equilibrium

THE CLASSICAL MACROECONOMIC MODEL

Friedman's arguments are best conveyed through the Classical or long-run macroeconomic model of income determination.

1

PRODUCTION FUNCTION

Output depends on employment of productive factors: labour, capital, and land.

2

VARIABLE FACTOR

Only labour can be quickly varied over the business cycle; capital is fixed at any given time.

3

OUTPUT DETERMINATION

Supply of output increases and decreases as the amount of labour employed changes.

LABOUR MARKET EQUILIBRIUM

What determines employment levels? The equilibrium of supply and demand in the labour market.

THE REAL WAGE

The ratio of the money wage to the price level—this is the relevant price of labour, not the money wage.

For workers: Measures purchasing power over goods and services.

For firms: Measures true cost of paying workers a given money wage.

MARKET DYNAMICS

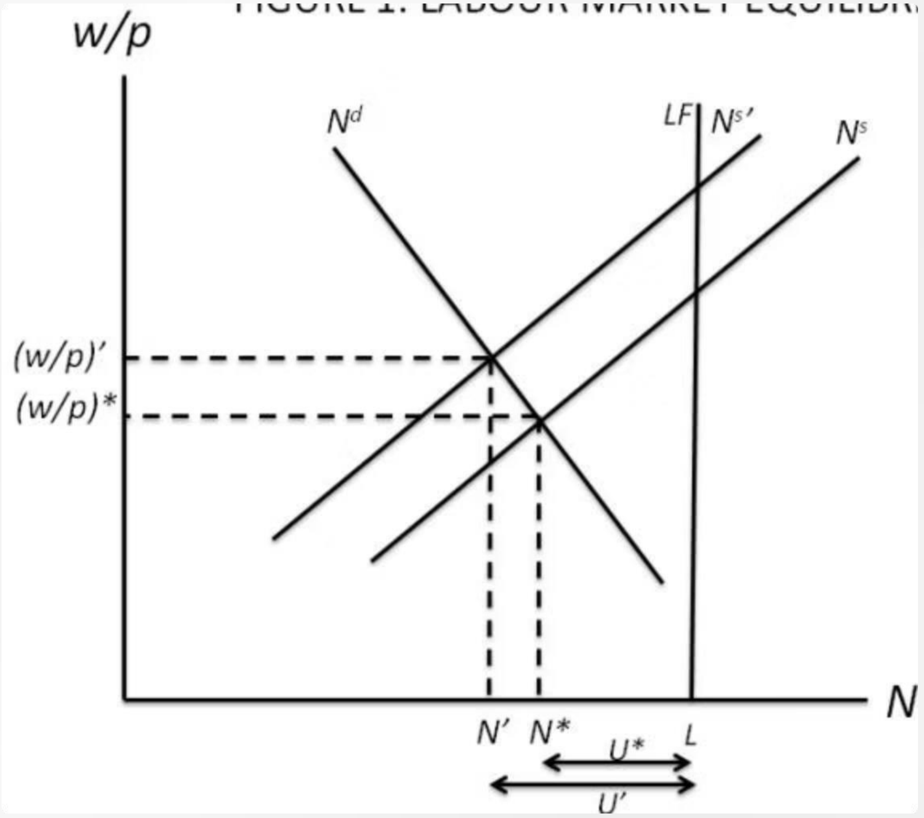
Supply: Increases with the real wage (greater reward for working).

Demand: Decreases with the real wage (increasing cost to employer).

Equilibrium: Where the labour market 'clears'.

FIGURE 1: LABOUR MARKET EQUILIBRIUM

The graph illustrates labour market equilibrium with supply curve Ns (upward-sloping), demand curve Nd (downward-sloping), and long-run aggregate supply LF (vertical). Initial equilibrium occurs at (N*, (w/p)*).



KEY EQUILIBRIUM CONCEPTS

MARKET CLEARING

Equilibrium at N*, (w/P)* where supply and demand intersect.

NATURAL EMPLOYMENT

Long-run employment level follows this equilibrium.

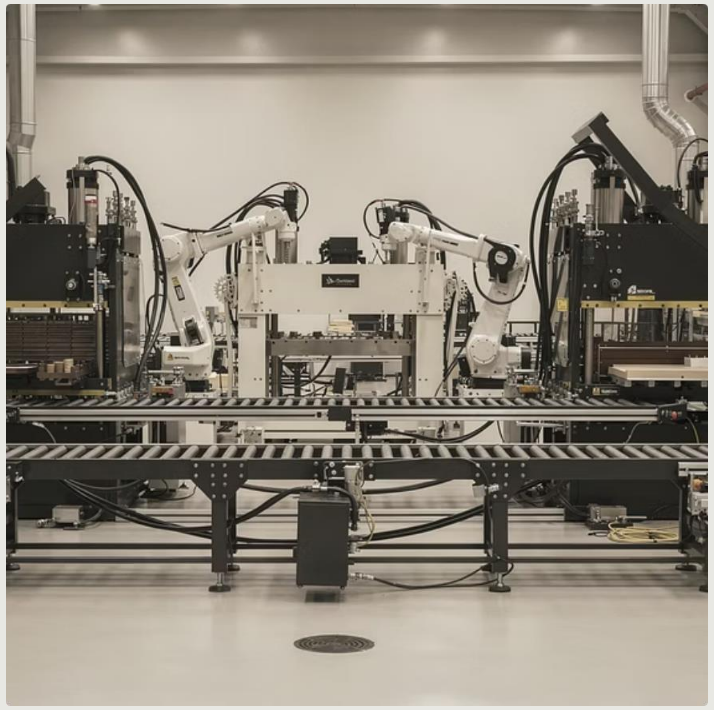
NATURAL UNEMPLOYMENT UNEMPLOYMENT RATE

$u^* = (L - N^*)/L = U^*/L$

Some unemployment exists at equilibrium (frictional unemployment), but this is consistent with market-clearing. Individual workers move in and out of this pool, so no one is seriously disadvantaged.

LONG-RUN UNEMPLOYMENT DYNAMICS

The 'long-run' rate of unemployment is not necessarily permanent over time. It depends on exogenous factors that change over longer periods:



CAPITAL STOCK

An increase shifts the Nd curve rightward, raising employment.

LEISURE PREFERENCES

Increased preference for leisure shifts labour supply curve leftward (shown in Figure 1).

PART II

DICHOTOMY AND NEUTRALITY

A macroeconomic model of money's role

KEY MACROECONOMIC VARIABLES

Following Friedman, we construct a macro model to determine these variables, demonstrating the dichotomy and neutrality of money:

- **GROSS DOMESTIC PRODUCT (GDP)**
- **UNEMPLOYMENT RATE**
- **PRICE LEVEL (CPI)**
- **INTEREST RATES**

THE LABOUR MARKET MODEL

Let w/P be the real wage, denoted by θ. The labour market can be represented by:

```
N^s = N^s(θ, ε)    N^s_θ > 0; N^s_ε < 0
N^d = N^d(θ, K)    N^d_θ < 0; N^d_K > 0
N^s = N^d
```

LABOUR SUPPLY

Upward-sloped function of real wage. Increase in ε (taste for leisure) shifts it leftward.

LABOUR DEMAND

Downward-sloped function of real wage. Increase in capital stock K shifts it rightward.

SOLVING THE LABOUR MARKET

EQUILIBRIUM CONDITION

NS = ND = N

SYSTEM OF EQUATIONS

Two equations, two unknowns: N and θ (ε and K are exogenous parameters)

UNIQUE SOLUTION

N* and θ* represent long-run equilibrium given exogenous parameters

OUTPUT DETERMINATION

Output supplied depends on employment of productive factors through an **aggregate production function**:

$$Y^s = F(K, N)$$

K IS FIXED	N IS DETERMINED	YS IS DETERMINED
Capital stock determined exogenously	Employment from labour market equilibrium	Output supply follows directly

Critical insight: The supply of output is purely determined by (i) labour market equilibrium and (ii) the aggregate production function, given the amount of capital.

THE REAL SECTOR

N and Y have been solved without examining the demand side of the goods market or the money market.

1

OUTPUT DETERMINED

Ys from supply side

2

DEMAND ADJUSTS

Yd must equal Ys

3

INTEREST RATE EQUILIBRATES EQUILIBRATES

i adjusts to ensure Yd = Y

What might affect Y? Changes in preference for leisure, capital stock, or other supply-side factors such as technological progress or population growth.

THE IS RELATION

Aggregate demand for goods can be represented as:

$$Y^d = Y^d(i, G, T) \quad Y_i < 0, Y_G > 0, Y_T < 0$$

VARIABLES

Yd: Aggregate demand

i: Real interest rate

G: Government spending

T: Tax revenues

RELATIONSHIPS

Positive: Government spending

Negative: Interest rate and taxes

EQUILIBRIUM

Yd must equal Ys (already determined)

Only i can adjust to ensure equilibrium

INTEREST RATE DETERMINATION

> The interest rate i is determined by equilibrium of aggregate demand and aggregate supply in the goods market, with the level of output Y already determined from the supply side of the economy.

This is the Classical 'loanable funds' theory of the interest rate, where the interest rate ensures that S = I (sum of household and government savings equals investment).

THE MONEY MARKET

All variables encountered so far are 'real', including interest rate i. We have made no reference to money supply M or price level P.

The demand for money is described by the **Quantity Theory of Money**:

$$M^d = k \quad P \quad Y$$

MEDIUM OF EXCHANGE

Money facilitates transactions

UNIT OF ACCOUNT

Determines nominal variables like P and w

SOLVING FOR THE PRICE LEVEL

Money supply Ms is exogenously set by the monetary authority. Equilibrium condition: Ms = Md = M

$$M = kPY$$

Since all endogenous variables except one are determined in the real sector, only the price level P remains:

$$P = \frac{M}{kY}$$

M: Set by central bank

k: Constant (velocity)

Y: Determined by supply side

Once P is known, money wages can be calculated: $w = \theta \times P$

THE CLASSICAL DICHOTOMY

REAL SECTOR DETERMINES

- Employment (N)
- Real wage (θ)
- Aggregate output (Y)
- Interest rate (i)

Determined by labour market, production function, and IS relation.

MONEY SECTOR DETERMINES

- Price level (P)
- Money wage (w)

Determined solely by the money market.

Neutrality of money: A proportionate increase in M causes P (and w) to rise in the same proportion, with no effect on any real variable.

THE AGGREGATE DEMAND CURVE

Inverting the QTM equation reveals aggregate demand:

$$Y = \frac{1}{k} \frac{M}{P} \equiv Y^d$$

Planned expenditures on goods are proportional to the *real* value of the money stock.

AD CURVE PROPERTIES

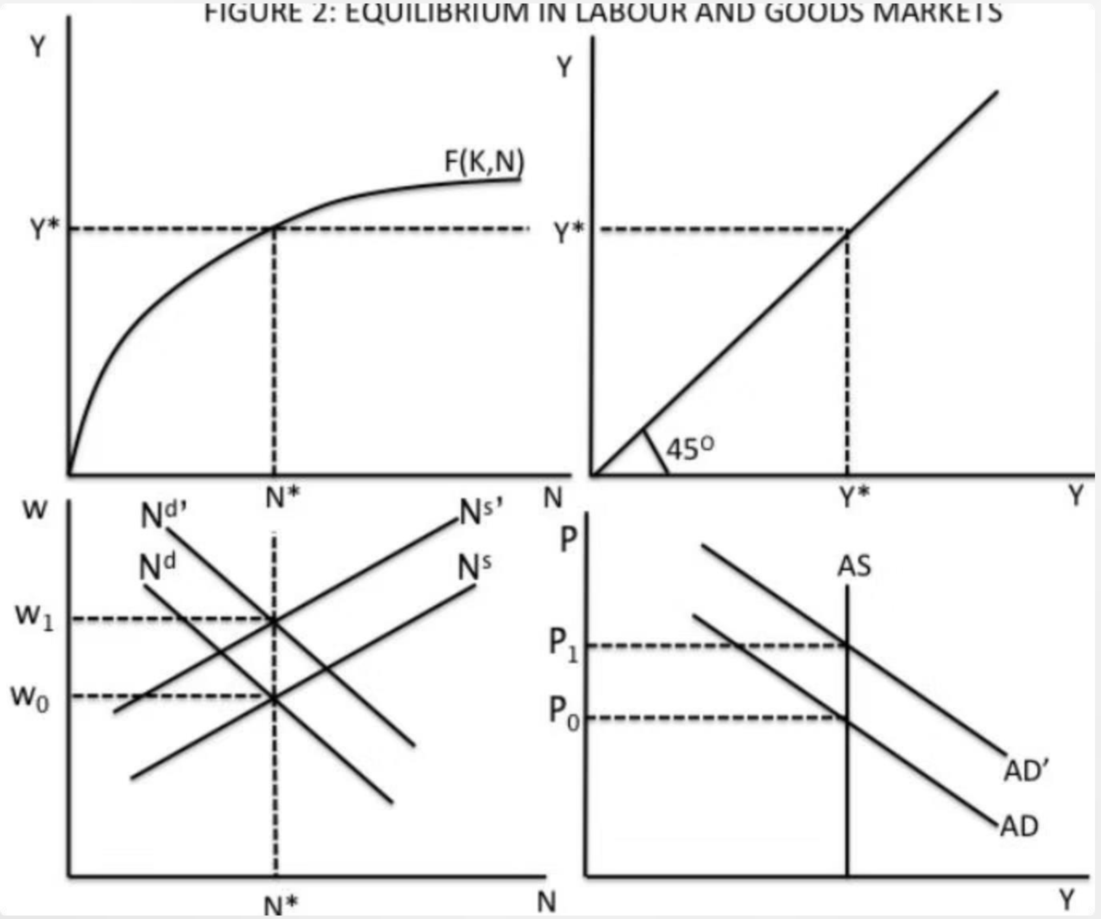
Downward sloping in the price level

MONETARY EXPANSION

Increases in M cause AD to shift outwards

FIGURE 2: EQUILIBRIUM EQUILIBRIUM IN LABOUR AND GOODS GOODS MARKETS

Four-panel diagram illustrating the determination of endogenous variables. Top-left: production function F(K,N). Top-right: 45-degree line. Bottom-left: labour market with Nd and Ns curves. Bottom-right: goods market with AS and AD curves.



KEY PRINCIPLE

MONEY IS NEUTRAL NEUTRAL

In the macroeconomic model with labour market clearing, changes in the money supply have no lasting effect on real variables—only on nominal variables like prices and wages.

PART III

SHORT-TERM MONEY ILLUSION

Temporary departures from neutrality

FRIEDMAN'S SHORT-RUN ACKNOWLEDGEMENT

Friedman acknowledged that neutrality might not hold in the short run following an increase in the money supply.

> In the short run, workers may suffer temporary 'money illusion'—willing to supply more labour when observing higher money wages, even if their real wage hasn't changed or has decreased.

WHY MONEY ILLUSION OCCURS

WORKERS' PERSPECTIVE

Workers are paid in money, so they directly observe w (money wage).

They may not have a correct idea about their real wage, since knowing that requires knowing P (price level), which is not directly observable.

FIRMS' PERSPECTIVE

For an individual firm, the relevant real wage is W divided by the price of its **own** product.

This is directly observable by each firm, just as w is directly observable by each worker.

The macroeconomic real wage w/P only emerges when all firms are aggregated. Thus employers perceive changes in real wages more easily than employees do.

SHORT-RUN EFFECTS OF MONETARY EXPANSION

With short-run money illusion, an increase in P can lead to:

LABOUR DEMAND SHIFTS

Immediate rightward shift in labour demand curve, whilst labour supply remains stable

INCREASED LABOUR SUPPLY

Employees observe higher money wages and mistakenly believe real wages are rising, supplying more labour

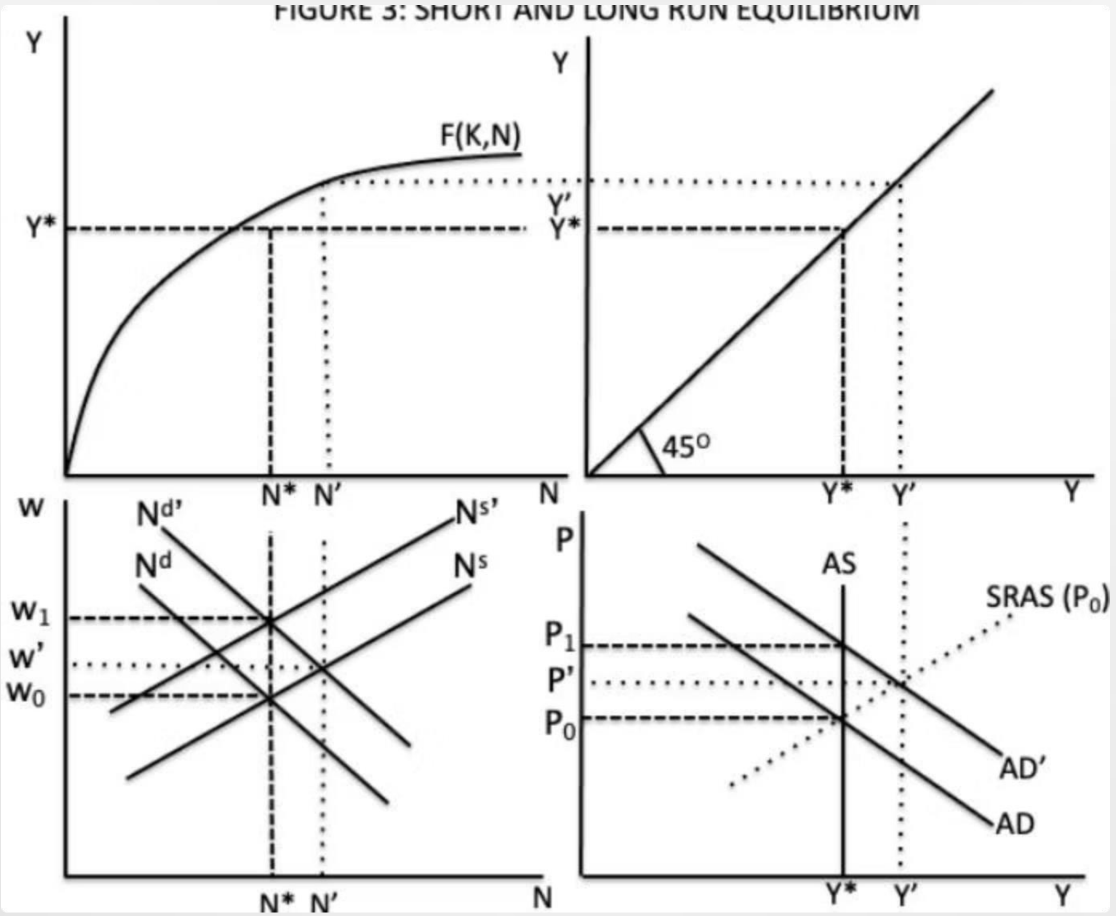
OUTPUT EXPANSION

More output is produced as employment increases

UPWARD-SLOPING AS

In the short run, the AS curve can be upward sloping

FIGURE 3: SHORT AND LONG RUN EQUILIBRIUM



Four-panel diagram showing departure of short-run equilibrium from long-run counterpart. Note the temporary increase to N' and Y' before returning to natural levels N* and Y*.

SHORT-RUN DYNAMICS

Following a shock to aggregate demand, output and employment may increase in the short run due to short-term money illusion on the part of labour.

LABOUR MARKET

Money wages increase but less than the price level; therefore real wages fall, incentivising firms to employ more workers.

TEMPORARY EQUILIBRIUM

New equilibrium at N', Y', P', and w' based on false perception that cannot last into the long run.

LONG-RUN ADJUSTMENT PROCESS

Eventually workers realise that despite higher money wages, the price level has risen even more. They begin withdrawing labour at existing wages.

1. **LABOUR SUPPLY SHIFTS**
   Labour supply curve shifts left to Nsl
2. **EMPLOYMENT RETURNS**
   Employment returns to natural level N*
3. **WAGES RISE FURTHER**
   Money wages rise to w_1
4. **OUTPUT NORMALISES**
   Output returns to natural level Y*
5. **PRICES INCREASE**
   New, higher price level P_1 proportionate to higher money supply
6. **REAL WAGES RESTORED**
   Real wages return to $\theta^* = w_1/P_1 = w_0/P_0$

INTEREST RATE DYNAMICS

SHORT RUN

With Y' from the analysis, the IS relation determines short-run interest rate i':

$$Y' = Y(i', G, T)$$

Since Y' > Y* and Y and i are negatively related along IS, we conclude i' < i*.

LONG RUN

The IS relation dictates:

$$Y \quad = \quad Y(i \;, G, T)$$

where i* is the long-run 'natural' interest rate for given G and T.

Summary: In the short run, monetary expansion causes interest rates to fall. In the long run, interest rates return to the natural level.

FRIEDMAN'S POLICY WARNING

Whilst acknowledging short-run effects, Friedman argued strongly *against* using monetary policy to lower unemployment or stimulate production:

TEMPORARY EFFECTS

Effects on unemployment are temporary; price levels ultimately bear the brunt of monetary expansion.

POLICY LAGS

Monetary policy operates with lags, so attempts to address real sector problems likely end in excessive expansion and unduly high inflation.

REVERSAL PROBLEMS

When policy reverses, excessive contraction causes disinflation or deflation.

BEST PRACTICE

The best monetary policy keeps inflation moderate and stable.

PART IV

SUPER-NEUTRALITY OF MONEY

Dynamic monetary growth and inflation

FROM ONE-OFF CHANGES TO SUSTAINED GROWTH GROWTH

We have studied what happens following a single monetary expansion—a short increase in output and employment, with lasting effect of raising the price level proportionately.

1

ONE-OFF INCREASE

Not the same as inflation

2

INFLATION

Sustained increases over time

3

SYSTEMATIC GROWTH

Requires ongoing money supply growth

For ongoing inflation, we must examine systematic growth in the money supply.

DYNAMIC QUANTITY THEORY OF MONEY

A dynamic version of the QTM:

$$\frac{\Delta M}{M} = \frac{\Delta P}{P} + \frac{\Delta Y}{Y}$$

ΔM/M

Rate of increase (or decrease) in money supply

ΔP/P

Rate of inflation

ΔY/Y

Growth rate of output

When $\Delta M/M > 0$ over a sustained period, the money supply will steadily grow. Velocity of money is constant ($\Delta V = 0$).

OUTPUT GROWTH AND INFLATION

If technological progress or capital accumulation occurs on the supply side, output will grow steadily ($\Delta Y/Y > 0$) even if total employment is constant.

Rearranging the dynamic QTM:

$$\frac{\Delta P}{P} = \frac{\Delta M}{M} - \frac{\Delta Y}{Y}$$

> The rate of inflation equals the growth rate of the money supply minus the growth rate of output.

LONG-RUN INFLATION DYNAMICS

In the long run, the labour market is at its natural level of employment (constant). Thus the growth rate of output is at its natural rate.

Any *change* in the growth rate of money is passed on to *changes* in the rate of inflation:

$$\frac{\Delta P'}{P} = \frac{\Delta M'}{M} - \frac{\Delta Y}{Y}$$

where ' indicates new growth rates of money and inflation, given constant output growth.

FAMOUS DICTUM

INFLATION IS ALWAYS AND EVERYWHERE A MONETARY PHENOMENON

—Milton Friedman

Inflation can only be produced by a more rapid increase in the quantity of money than in output.

SIMPLIFIED LONG-RUN RELATIONSHIP

Assuming the rate of economic growth is zero, inflation in the long run equals the growth rate of money:

$$\pi = g^m$$

where π represents the rate of inflation and gm is the rate of growth of the money supply.

Question: Suppose an economy is in long-run equilibrium when the monetary authority accelerates the growth rate of money. What is the short-run impact on macroeconomic variables?

SHORT-RUN EFFECTS OF ACCELERATED MONEY GROWTH GROWTH

Assuming money illusion on the part of employees, an increase in gm leads to:

INFLATION ACCELERATES

But not at the same rate as money is growing

REAL WAGES FALL

Employees don't perceive higher inflation, accept below-inflation wage increases

EMPLOYMENT RISES

Workers supply more labour; firms employ more due to lower real wages

OUTPUT INCREASES

Temporary increase in production

Actual π lags behind gm; N and Y both increase as actual decrease in real wages holds costs down.

LONG-RUN ADJUSTMENT TO SUSTAINED GROWTH

Workers' perceptions of actual inflation and its effect on real wages eventually catch up:

LABOUR SUPPLY FALLS

Workers withdraw labour as they perceive falling real wages

OUTPUT AND EMPLOYMENT NORMALISE

Return towards natural levels

REAL WAGES RISE

Return to long-run equilibrium levels

INFLATION CATCHES UP

Rate of inflation rises to equal the higher gm

IRVING FISHER (1867–1947)

American economist and mathematician famous for contributions to general equilibrium theory and capital theory.

Developed the relationship between nominal interest rates, real interest rates, and inflation—the **Fisher equation**.

THE FISHER EQUATION

With ongoing inflation, we must distinguish between nominal and real interest rates:

$$i = r + \pi$$

where r is the real interest rate and i is the nominal interest rate.

EX POST FORM

$$r = i - \pi$$

Real return equals nominal interest minus actual inflation between purchase and maturity.

EX ANTE FORM

$$i = r + \pi^e$$

To achieve real return r, nominal rate should equal r plus expected inflation.

Special case: When $\pi = \pi e = 0$, then $i = r$ (nominal equals real interest rate).

INTEREST RATE EFFECTS OF MONEY GROWTH

LONG RUN

r is a real sector variable independent of money and its growth rate.

Higher gm ⇒ higher π ⇒ higher πe ⇒ higher i

If money growth rate increases, long-run effect is to increase nominal interest rate charged by lenders.

SHORT RUN

As gm increases:

- Ex post real interest rate on previous loans falls as inflation rises
- Lenders don't perceive higher inflation, so πe stays at original π
- Ex ante nominal interest rate falls due to competitive pressure

Both nominal and real rates fall initially.

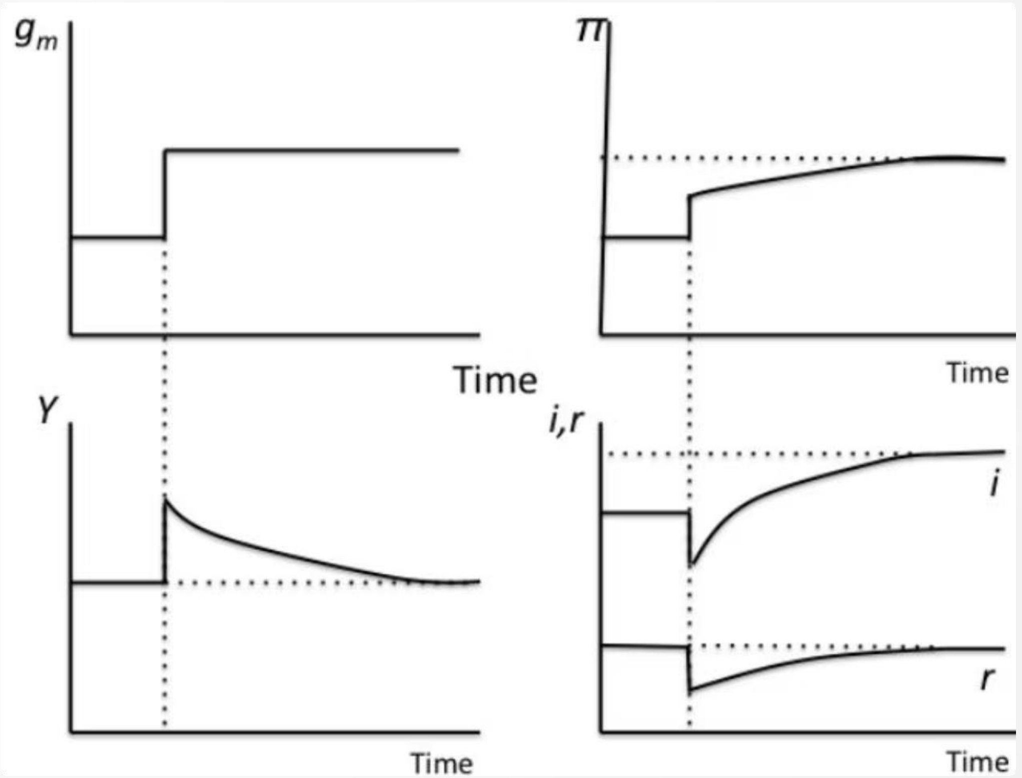


FIGURE 4: DYNAMIC EFFECTS OF MONETARY EXPANSION

Four graphs showing time-series effects: gm (money growth) steps up permanently; M (money supply) rises; Y (output) rises temporarily then falls back; i and r (interest rates) rise temporarily then settle at new levels.

SUMMARY

POLICY CONCLUSIONS

01

MONEY'S IMPORTANCE

Important influence on aggregate demand and macroeconomic equilibrium

02

NEUTRALITY

Changes in money supply don't have lasting effects on real variables, but permanent effects on nominal variables

03

SUPER-NEUTRALITY

Percentage point increase in money growth rate increases inflation and nominal interest rates by same amount

04

SHORT-RUN EFFECTS

Money illusion allows transitory effects on real variables, but this doesn't justify using monetary policy to stimulate activity

05

FRIEDMAN'S K% RULE

Increase money supply at moderate, steady pace to accommodate liquidity needs of growing economy

06

PRIMARY OBJECTIVE

Promote price stability through monetary stability—don't use policy to 'fine tune' the economy